

Prakash Chand Palsaniya

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OBJECTIVE

IT student at NIT Srinagar with an 8.4 CGPA and a strong interest in data engineering and backend development. I enjoy solving problems, learning new technologies, and building reliable systems. Solved 500+ DSA problems and have a good foundation in programming, databases, and software development. Looking forward to applying these skills to real-world data and engineering problems.

SKILLS

Languages: C++, C, Python, JavaScript, SQL
AI / LLM: LangChain, LangGraph, RAG, Qdrant
Backend: Node.js, Express.js, REST APIs, Microservices, RabbitMQ
Databases: PostgreSQL, MongoDB, MySQL, Redis
Cloud & Tools: AWS (EC2, S3, Lambda, CloudFront), Docker, Git, Postman
Core CS: Data Structures & Algorithms, OOP, DBMS

EDUCATION

National Institute of Technology, Srinagar
B.Tech. in Information Technology (CGPA: 8.4)

Srinagar, India
Aug. 2023 – Aug. 2027

PROJECTS

DocuMind | RAG Document Intelligence Platform
LangChain, LangGraph, Qdrant, PostgreSQL, Docker

[Live Demo](#) | [GitHub](#)

- Built an AI document assistant that answers questions from uploaded PDFs with source citations, letting users find answers without reading entire documents.
- Designed the agent as a LangGraph state machine (query rewriting, relevance grading, self-correcting retry, web-search fallback), so weak retrieval is retried instead of answered with a guess.
- Engineered the retrieval pipeline (parsing, chunking, embedding, hybrid BM25 + vector search with cross-encoder reranking), significantly improving answer accuracy and reducing incorrect responses.
- Implemented semantic caching that reuses answers to meaning-equivalent questions, cutting LLM API cost and returning repeat queries almost instantly.
- Created an automated LLM-as-judge evaluation harness scoring answer faithfulness and relevance, making response quality measurable and improvements data-driven.

FinMate | AI-Powered Personal Finance Analytics Platform
Node.js, MongoDB, Express, Redis, Gemini AI, AWS EC2, Docker

[Live Demo](#) | [GitHub](#)

- Integrated Gemini AI as a finance assistant delivering personalized spending insights and letting users query transactions in plain English.
- Shipped a Telegram bot that logs expenses from everyday chat messages, removing app friction and driving faster, more consistent expense tracking.
- Built complex MongoDB aggregation pipelines for spending trends and top expense categories, cutting dashboard load time by 85% (800ms to 120ms) with Redis caching; deployed on AWS EC2 using Docker.
- Implemented secure multi-method login with Google sign-in, email OTP, and rate limiting to protect accounts against brute-force attacks.

PKLinks | Distributed URL Shortener & Click Analytics System
Node.js, Express.js, MongoDB, Redis, RabbitMQ, Docker

[Live Demo](#) | [GitHub](#)

- Built a URL shortener and analytics platform on a 6-service microservices architecture behind an API Gateway, allowing each service to scale and deploy independently.
- Cut redirect latency by 60% using Redis caching, keeping links fast for end users even under high traffic volume.
- Designed an asynchronous RabbitMQ analytics pipeline that tracks click location and device data without slowing redirects, feeding a dashboard of click trends, traffic sources, and geographic and device breakdowns.
- Developed JWT authentication with registration, login, and secure password reset, with emails delivered by a separate Notification Service.